JABchem

Past Papers Higher Chemistry

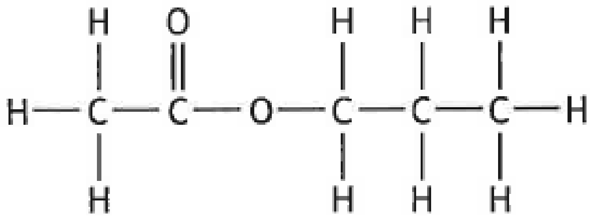
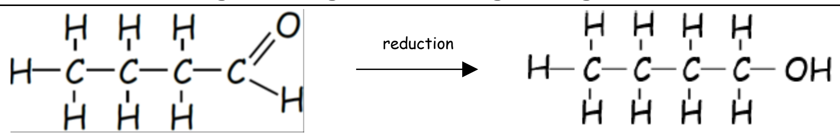
2025

Marking Scheme

Grade Awarded	Mark Required		% candidates achieving grade
	(/150)	%	
A	108+	72%	32.0%
B	92+	61%	25.9%
C	77+	51%	18.2%
D	61+	41%	13.3%
No award	<61	<41%	10.6%

Section:	Multiple Choice	Extended Answer	Assignment
Average Mark:	17.6 /25	56.5 /95	19.7 /30

2025 Higher Chemistry Marking Scheme

MC Qu	Answer	% Pupils Correct	Reasoning
1	A	87	☑A Electronegativity difference of N-Br bond = 3.0 – 3.0 = 0.0 (pure covalent) ☒B Electronegativity difference of Cl-H bond = 3.2 – 2.2 = 1.0 ☒C Electronegativity difference of O-H bond = 3.4 – 2.2 = 1.2 ☒D Electronegativity difference of I-C bond = 2.7 – 2.6 = 0.1
2	C	77	☒A Reducing agents are oxidised themselves and lose electrons ☒B Strong reducing agents are metals on top right of Electrochemical Series. Metals are found on left side of the Period Table ☑C Strong reducing agents are group 1+2 metals with low electronegativity values ☒D Strong reducing agents are metals which lose electrons to form positive ions
3	B	72	no. of moles = volume x concentration = 0.1 litres x 0.5 mol l^{-1} = 0.05mol $MnO_4^- + 5Fe^{2+} + 8H^+ \longrightarrow Mn^{2+} + 5Fe^{3+} + 4H_2O$ MnO_4^-: 1mol, 0.05mol $5Fe^{2+}$: 5mol, 0.25mol
4	D	53	**3,4-dimethylhex-3-ene** 3,4: Side-groups on C_3 and C_4 dimethyl: 2x $-CH_3$ methyl side groups hex: 6 Carbons in main chain 3: Functional Group on C_3 ene: C=C double bond
5	B	90	☒A Primary: 1 carbon directly attached to the carbon with the -OH group ☑B Tertiary: 3 carbons directly attached to the carbon with the -OH group ☒C Secondary: 2 carbons directly attached to the carbon with the -OH group ☒D Secondary: 2 carbons directly attached to the carbon with the -OH group
6	A	93	☑A Structure is 3-ethylhexanoic acid ☒B Structure is 2-ethylhexanoic acid ☒C Structure is 3,3-dimethylhexanoic acid ☒D Structure is 4,4-dimethylhexanoic acid
7	B	69	 Carboxylic Acid: ethanoic acid, gfm = 60g Alcohol: propan-1-ol, gfm 60g
8	D	64	 aldehyde butanal C_4H_8O gfm= 72g primary alcohol butan-1-ol $C_4H_{10}O$ gfm = 74g

9	C	74	$\underbrace{\text{6-methyl}}_{\text{methyl -CH}_3\text{ group on C}_6}\underbrace{\text{hept-}}_{\text{Seven carbons on main chain}}\underbrace{\text{5-ene-}}_{\text{C=C double bond between C}_5\text{ and C}_6}\underbrace{\text{2-one}}_{\text{C=O Carbonyl group on C}_2}$ ❼ ❻ ❺ ❹ ❸ ❷ ❶ $CH_3\underbrace{C\overbrace{(CH_3)}^{\text{methyl -CH}_3\text{ group on C}_6}CH}_{\text{C=C double bond between C}_5\text{ and C}_6}CH_2CH_2\underbrace{CO}_{\text{C=O Carbonyl group on C}_2}CH_3$
10	B	72	☒A hydration: water added across a C=C double bond ☑B fat is hydrolysed into glycerol and 3 fatty acids. Fatty acids then neutralised. ☒C esterification: alcohol and carboxylic acid join together with H_2O removed ☒D condensation: two molecules join together with small molecule e.g. H_2O removed
11	D	82	☒A fatty acid chains are hydrocarbon ∴ hydrophobic and insoluble in water ☒B fatty acid chains are hydrocarbon ∴ hydrophobic and insoluble in water ☒C fatty acid chains are hydrocarbon ∴ hydrophobic ☑D fatty acid chains are hydrocarbon ∴ hydrophobic and soluble in oil droplets
12	B	84	☒A terpenes must have a formula with a multiple of five carbons ☑B Terpene with formula $C_{15}H_{24}$ is made from three C_5H_8 isoprene joined together ☒C terpenes must have a formula with a multiple of five carbons ☒D terpenes must have a formula with a multiple of five carbons
13	C	79	When rate = 0.02 s^{-1} — Temperature = 44°C When rate = 0.04 s^{-1} — Temperature = 56°C ∴ Change in temperature = 12°C
14	A	71	☑A Line T_2 is to the right and not as high as line T_1 ☒B The area under the curves should be the same for both lines ☒C Line T_2 should be lower than line T_1 ☒D Line T_2 should be to right of line T_1
15	A	64	Enthalpy Change = activation energy for Forward Reaction - activation energy for Reverse Reaction Enthalpy Change = 165 - 179 Enthalpy Change = -14 kJ
16	D	57	☒A High temperature will favour the endothermic reverse reaction reducing B ☒B High activation energy decreases number of particles forming activated complex ☒C High temperature will favour the endothermic reverse reaction reducing B ☑D Low temperature favours forward reaction & low activation energy favours B formation
17	D	78	c $N_2O_4 \rightarrow 2NO_2$ a $2NO + O_2 \rightarrow N_2O_4$ b $2NO + O_2 \rightarrow 2NO_2$ -a $N_2O_4 \rightarrow \cancel{2NO} + \cancel{O_2}$ b $\cancel{2NO} + \cancel{O_2} \rightarrow 2NO_2$ c = -a +b $N_2O_4 \rightarrow 2NO_2$
18	B	74	❸ $2CH_4 + 3O_2 \rightarrow 2CO + 4H_2O$ ❶ $CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$ ❷ $2CO + O_2 \rightarrow 2CO_2$ ❶x2 $2CH_4 + {}^{3}\cancel{4}\ O_2 \rightarrow \cancel{2CO_2} + 4H_2O$ ❷x-1 $\cancel{2CO_2} \rightarrow 2CO + \cancel{O_2}$ ❸ $2CH_4 + 3O_2 \rightarrow 2CO + 4H_2O$

19	D	82	☒A catalysts lower the activation energy ☒B adding a catalyst does not alter the kinetic energy of the reactants ☒C catalyst do not alter the position of equilibrium ☑D catalyst have no effect on the enthalpy change of a reaction
20	C	46	$\underset{100\%}{A} \longrightarrow \underset{50\%}{B} \longrightarrow \underset{40\%}{C}$ $= 80\%$ of 50% $= \frac{80}{100} \times 50\%$ $= 40\%$
21	D	43	Formula of Aluminium Phosphate = $AlPO_4$ Formula of Aluminium Sulphate = $Al_2(PO_4)_3$ $AlPO_4$ $Al_2(PO_4)_3$ 1mol : 1mol 2mol : 3mol $\underbrace{\text{1mol : 1mol} \quad \text{4mol : }\textbf{6mol}}_{\text{5 mol } Al^{3+} \text{ ions}}$
22	A	62	☑A One mole of electrons is added to one mole of atoms in the gaseous atoms ☒B Bond in Cl_2 must be broken into Cl atoms before electron affinity takes place ☒C Bond in Cl_2 must be broken into Cl atoms before electron affinity takes place ☒D This equation is the first ionisation energy where positive ions are formed
23	C	36	☒A Adding HCl increases concentration of product H^+ ∴ equilibrium moves to left ☒B Adding KBr increases concentration of product Br^- ∴ equilibrium moves to left ☑C Adding $AgNO_3$ causes precipitation of $AgBr_{(s)}$. This reduces the concentration of product Br^- ∴ equilibrium moves to right ☒D Adding NaOCl increases concentration of product OBr^- ∴ equilibrium moves to left
24	C	62	☒A The last few drops should be added drop by drop so the colour change can be calculated to the nearest drop. This improves the accuracy of the titration. ☒B Using a white tile allows the colour change to be determined exactly. This improves the accuracy of the titration. ☑C Using a dry conical flask is not essential as number of moles of reactant in the conical flask is decided by the volume and concentration of the reactant added. ☒D Swirling the flask mixes the solutions better and ensures the colour change is accurate to the volume in flask. This improves the accuracy of the titration.
25	A	69	Starting volume = $12.5cm^3$ End volume = $24.3cm^3$ Change in volume = $11.8cm^3$

2025 Higher Chemistry Marking Scheme

<table>
<tr><th>Long Qu</th><th>Answer</th><th>Reasoning</th></tr>
<tr><td>1a(i)</td><td>$Na(g) \rightarrow Na^{+}(g) + e^{-}$</td><td>The first ionisation energy is the energy to remove one mole of electrons from one mole of atoms in the gaseous state.</td></tr>
<tr><td>1a(ii)</td><td>1735</td><td>❶ $Ca(g) \longrightarrow Ca^{+}(g) + e^{-}$ $\Delta H = 590kJ$
❷ $Ca^{+}(g) \longrightarrow Ca^{2+}(g) + e^{-}$ $\Delta H = 1145kJ$

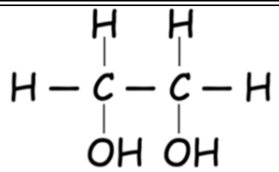
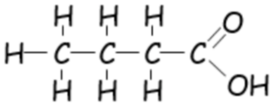
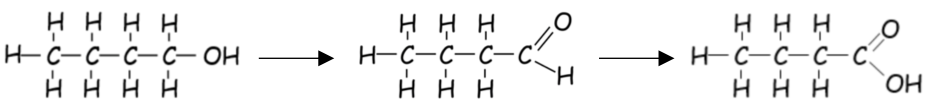
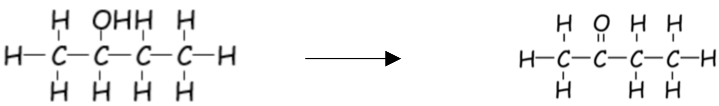
❶+❷ $Ca(g) \longrightarrow \cancel{Ca^{+}(g)} + e^{-}$ $\Delta H = 590kJ$
$\cancel{Ca^{+}(g)} \longrightarrow Ca^{2+}(g) + e^{-}$ $\Delta H = 1145kJ$

$Ca(g) \longrightarrow Ca^{2+}(g) + 2e^{-}$ $\Delta H = 1735kJ$</td></tr>
<tr><td>1a(iii)</td><td>Answer to include:</td><td>
<table>
<tr><td rowspan="2">1st mark:
(one from)</td><td>The second ionisation energy removes an electron from an electron shell that is</td><td>inner
full
(more) stable
closer</td><td>to the nucleus</td></tr>
<tr><td>Second electron removed from a shell that is</td><td>inner
full
(more) stable
closer</td><td>to the nucleus</td></tr>
</table>
<table>
<tr><td rowspan="2">2nd mark:
(one from)</td><td>The second</td><td>electron
electron shell</td><td>is less</td><td>screened
shielded</td></tr>
<tr><td>Second electron is more strongly</td><td>attracted to
pulled towards</td><td colspan="2">the nucleus</td></tr>
</table>
</td></tr>
<tr><td>1a(iv)</td><td>greater nuclear charge or more protons</td><td>The larger number of protons in the nucleus of Mg^{2+} have a greater pull on the outer shell than the Na^{+} nucleus has on its outer shell making the Mg^{2+} ion smaller than the Na^{+}</td></tr>
<tr><td>1b(i)</td><td>Electrostatic attraction between positive ions and negative ions</td><td>Ionic bonding is usually between metals and non-metals where metals form positive ions and non-metals form negative ions.
An ionic lattice is a three-dimensional structure of alternating positive and negative ions. Electrostatic attraction between positive and negative ions is strong and the ionic lattice is a solid with a high melting point.</td></tr>
<tr><td>1b(ii)</td><td>6</td><td>mass = no. of moles x gfm = 0.0012 x 39.1 = 0.0469g = 46.9mg

1 tablet = 46.9mg ∴ 8 tablets = 46.9mg x 8 = 375mg

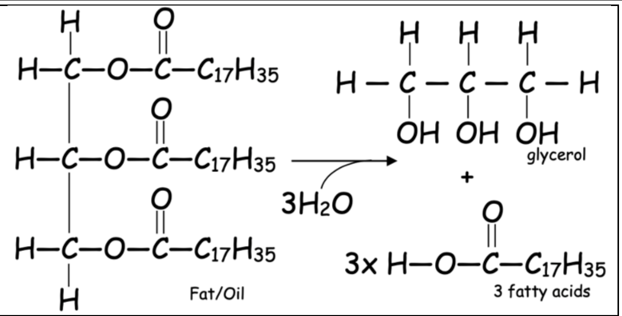
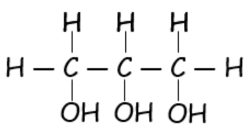
8 tablets for 1 day = 375mg ∴ 8 tablets for 7 days = 375mg x 8 = 2628mg

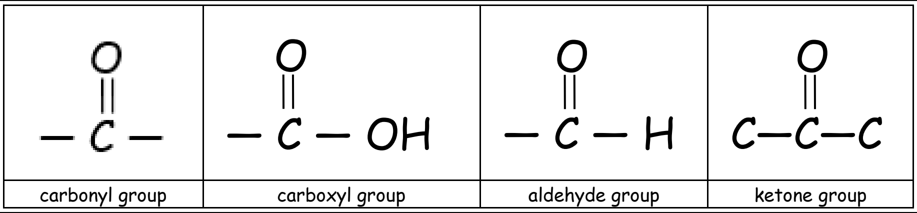
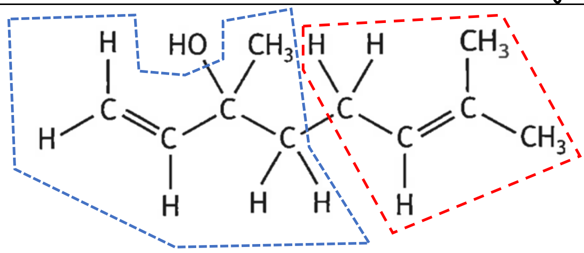
450mg K ⟷ 1 banana
2628mg K ⟷ 1 banana x $^{2628}/_{450}$
= 5.84 bananas
= 6 whole bananas</td></tr>
<tr><td>2a</td><td>
<table>
<tr><td>metallic</td><td></td></tr>
<tr><td></td><td>network</td></tr>
<tr><td>covalent</td><td></td></tr>
<tr><td></td><td>molecular</td></tr>
</table>
</td><td>
<table>
<tr><th>Bonding</th><th>Structure</th><th colspan="3">Elements in 3rd Period</th></tr>
<tr><td>Metallic</td><td>Lattice</td><td>Sodium Na</td><td>Magnesium Mg</td><td>Aluminium Al</td></tr>
<tr><td>Covalent</td><td>Molecular</td><td>Phosphorus P_4</td><td>Sulphur S_8</td><td>Chlorine Cl_2</td></tr>
<tr><td>Covalent</td><td>Network</td><td colspan="3">Silicon Si</td></tr>
<tr><td></td><td>Monatomic</td><td colspan="3">Argon Ar</td></tr>
</table>
</td></tr>
<tr><td>2b(i)</td><td>decreases</td><td>Going across the third period there are a greater number of protons in the nuclei in the elements and the greater nuclear charge pulls the outer shell more which reduces the covalent radius/atomic size.</td></tr>
</table>

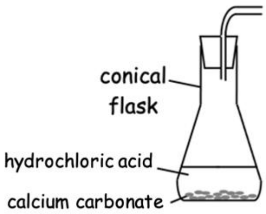
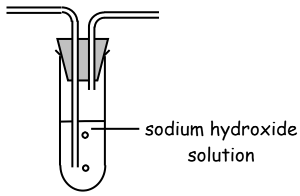
2b(ii)	Argon does not form (covalent) bonds	Group 0 elements like Argon have a full outer shell so do not need to share electrons and form covalent bonds to attain a full outer shell.
2c	Answer to include:	1st mark: (van der Waals/intermolecular) forces increase going down group 2nd mark: London dispersion forces are forces broken between atoms 3rd mark: The more electrons the stronger the forces
3	Open Question Answer: Mark: 0 / 1 / 2 / 3 %: 26% / 38% / 27% / 9%	**3 mark answer**: Demonstrates a **good understanding** of the chemistry involved. A good comprehension of the chemistry has provided in a logically correct, including a statement of the principles involved and the application of these to respond to the problem. **2 mark answer**: Demonstrates a **reasonable understanding** of the chemistry involved, making some statement(s) which are relevant to the situation, showing that the problem is understood. **1 mark answer**: Demonstrates a **limited understanding** of the chemistry involved. The candidate has made some statement(s) which are relevant to the situation, showing that at least a little of the chemistry within the problem is understood.
4a(i)	$C_3H_7OH + 4\frac{1}{2}H_2O$ ↓ $3CO_2 + 4H_2O$	$C_3H_7OH + 4\frac{1}{2}H_2O \longrightarrow 3CO_2 + 4H_2O$ (or any multiple)
4a(ii)	-1050	$E_h = c\ m\ T = 4.18 \times 0.15 \times 13.9 = 8.72$ kJ 0.498g propan-1-ol ⟷ -8.72 kJ 60g propan-1-ol ⟷ $-8.72 \text{ kJ} \times \frac{60}{0.498}$ $= -1050 \text{ kJ mol}^{-1}$
4a(iii)	Answers from:	Any four = 2 marks; Any two – 1 mark copper/metal can; thermometer; clamp stand; measuring cylinder; spirit burner & cap; balance
4a(iv)	One answer from	heat loss from surroundings; incomplete combustion; Evaporation of alcohol/fuel
4b(i)	1 : 3	 Formula: $C_2H_6O_2$ O : H 2 : 6 1 : 3
4b(ii)A	Acidified dichromate or hot copper (II) oxide	see table below
4b(ii)B		primary alcohol → aldehyde → carboxylic acid butan-1-ol → butanal → butanoic acid  Does ~~not~~ turn universal indicator red (butanal: Does not turn universal indicator red); (butanoic acid: Does turn universal indicator red)
4b(ii)C	butanone (accept butan-2-one)	secondary alcohol → ketone butan-2-ol → butanone  $C_4H_9OH \longrightarrow C_4H_8O$
4c(i)	Volumetric flask	Volumetric flasks (also known as standard flasks) are used to measure exact volumes of liquids with a mark on the narrow neck to fill up to.
4c(ii)A	Rinse with sodium hydroxide solution	Rinsing with sodium hydroxide solution and emptying the solution will ensure previous chemicals do not contaminate the sodium hydroxide solution and any water in the burette from previous rinsing does not dilute the sodium hydroxide.

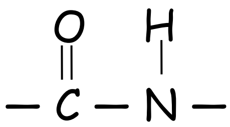
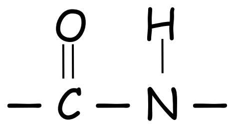
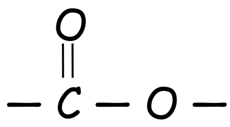
4b(ii)A table:

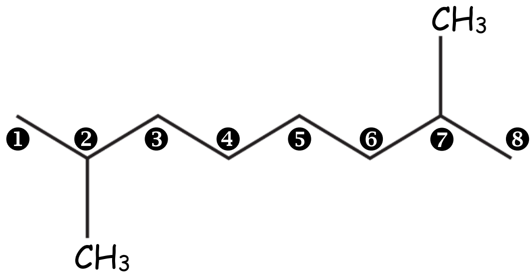
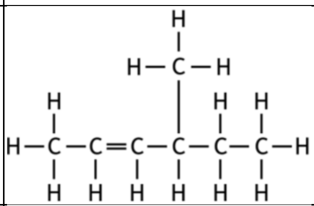
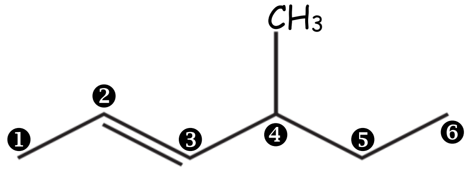
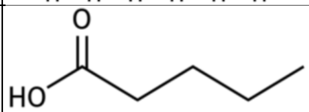
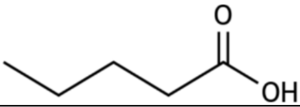
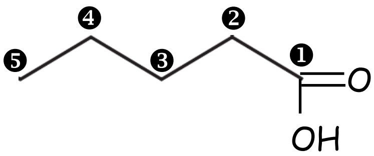
Oxidising Agent	Start Colour	Final Colour	Primary Alcohol ↓ Aldehyde	Secondary Alcohol ↓ Ketone	Aldehyde ↓ Carboxylic acid
Acidified dichromate	orange	green	✓	✓	✓
hot copper (II) oxide	black	brown	✓	✓	✓
Fehling's Solution	blue	brick red	×	×	✓
Tollen's Reagent	(colourless)	silver mirror	×	×	✓

<table>
<tr><td>4c(ii)B</td><td>20.4</td><td>average titre $= \frac{20.5 + 20.3}{2} = 20.4cm^3$</td></tr>
<tr><td>4c(ii)C</td><td>0.083</td><td>NaOH no. of moles = volume x concentration = 0.0198 litres x 0.105 mol l^{-1} = 0.002079mol
<table><tr><td>NaOH</td><td>:</td><td>CH_3COOH</td></tr><tr><td>1mol</td><td></td><td>1mol</td></tr><tr><td>0.002079mol</td><td></td><td>0.002079mol</td></tr></table>concentration $= \frac{\text{no. of moles}}{\text{volume}} = \frac{0.002079 \text{ mol}}{0.025 \text{ litres}} = 0.083 \text{mol } l^{-1}$</td></tr>
<tr><td>4c(iii)A</td><td>$K^+CH_3COO^-$</td><td>Potassium it is in group 1 of Periodic Table ∴ forms a positive ion with a 1+ charge.
Ethanoate has the formula CH_3COO^- (p9 of data booklet)</td></tr>
<tr><td>4c(iii)B</td><td>5.4</td><td>no. of moles = volume x concentration = 0.2 litres x 0.45mol l^{-1} = 0.09mol
mass = no. of moles x gfm = 0.09 x 60 = 5.4g</td></tr>
<tr><td>5a</td><td>Answer to contain:</td><td><table><tr><td>1st mark:</td><td>Sucrose is polar
due to hydroxyl groups</td><td>or</td><td>Sucrose can form hydrogen bonds
due to hydroxyl groups</td></tr><tr><td>2nd mark:</td><td colspan="3">polar substances like sucrose will dissolve in polar solvents like water</td></tr></table></td></tr>
<tr><td>5b(i)</td><td>Iso | Pro | Gly | Pro | Phe | Pro | Tyr
or
Tyr | Pro | Phe | Pro | Gly | Pro | Iso</td><td>There are 7 amino acids in the sequence to be worked out but 10 amino acids across the three smaller peptide chains ∴ three amino acids are ‘doublers’
(remember the chain could be written backwards)<table><tr><th>Look for common amino acids</th><th>Reverse Chain One</th><th>Line Up Common Fragments</th></tr><tr><td>Tyr-Pro-Phe</td><td>Phe-Pro-Tyr</td><td>Phe-Pro-Tyr</td></tr><tr><td>Gly-Pro-Phe</td><td>Gly-Pro-Phe</td><td>Gly-Pro-Phe</td></tr><tr><td>Iso-Pro-Gly-Pro</td><td>Iso-Pro-Gly-Pro</td><td>Iso-Pro-Gly-Pro</td></tr></table>Amino Acid Sequence = Iso – Pro – Gly - Pro- Phe – Pro - Tyr</td></tr>
<tr><td>5b(ii)</td><td>Essential</td><td>Essential amino acids are amino acids which the body cannot make for itself and must be obtained by the body through their dirt.</td></tr>
<tr><td>5c(i)</td><td>Esters
(or triglyceride)</td><td>Fats and oils are made from the condensation reaction between glycerol and three fatty acids. This makes fats and oils which are esters.
</td></tr>
<tr><td>5c(ii)A</td><td>propane-1,2,3-triol
or
glycerol</td><td>glycerol
$\underbrace{\text{prop}}_{\text{3 carbons}}\underbrace{\text{ane}}_{\text{Single bonds between carbons}}\text{-}\underbrace{\text{1,2,3}}_{\text{Functional groups on Carbons 1,2,3}}\text{-}\underbrace{\text{triol}}_{\text{3 hydroxyl -OH groups}}$</td></tr>
<tr><td>5c(ii)B</td><td>Palmitic Acid</td><td><table><tr><th>Fatty Acid</th><th>Formula</th><th>Fits General Formula</th><th>Number of C=C Double Bonds</th></tr><tr><td>Palmitic Acid</td><td>$C_{15}H_{31}COOH$</td><td>$C_nH_{2n+1}COOH$</td><td>0</td></tr><tr><td>Oleic Acid</td><td>$C_{17}H_{33}COOH$</td><td>$C_nH_{2n-1}COOH$</td><td>1</td></tr><tr><td>Linoleic Acid</td><td>$C_{17}H_{31}COOH$</td><td>$C_nH_{2n-3}COOH$</td><td>2</td></tr><tr><td>Linolenic Acid</td><td>$C_{17}H_{29}COOH$</td><td>$C_nH_{2n-5}COOH$</td><td>3</td></tr></table></td></tr>
<tr><td>5c(iii)</td><td>Addition
(accept reduction)</td><td>An addition reaction happens when a molecule adds across a C=C double bond.
an addition reaction where hydrogen is added across a C=C double bond is called hydrogenation
an addition reaction where water is added across a C=C double bond is called hydration</td></tr>
</table>

5d(i)	ethanal	
5d(ii)	carbonyl	
5d(iii)A	Diterpenoid	Terpene/Terpenoid molecules are made from isoprene C_5H_8 units joining together. Cafestol $C_{20}H_{28}O_3$ is made from four C_5H_8 units joining together.
5d(iii)B	One circled structure from:	 NB – OH group is not part of an isoprene unit as isoprene is a hydrocarbon
5e	3	1kg body mass $\longleftrightarrow$ 5.7mg caffeine 66kg body mass $\longleftrightarrow$ 5.7mg caffeine $\times \frac{66}{1}$ = 376.2mg caffeine 125.9mg caffeine $\longleftrightarrow$ 1 cup latte 376.2mg caffeine $\longleftrightarrow$ 1 cup latte $\times \frac{376.2}{125.0}$ = 3.01 cups latte = 3 cups maximum
6a	$H_2O_2 + 2H^+ + 2e^-$ ↓ $2H_2O$	redox $H_2O_2 + 2H^+ + 2I^- \rightarrow I_2 + 2H_2O$ oxidation $2I^- \rightarrow I_2 + 2e^-$ reduction $H_2O_2 + 2H^+ + 2e^- \rightarrow 2H_2O$
6b	$2S_2O_3^{2-} + I_2$ ↓ $S_4O_6^{2-} + 2I^-$	oxidation $2S_2O_3^{2-} \rightarrow S_4O_6^{2-} + 2e^-$ reduction $I_2 + 2e^- \rightarrow 2I^-$ add $2S_2O_3^{2-} + I_2 + \cancel{2e^-} \rightarrow 2I^- + S_4O_6^{2-} + \cancel{2e^-}$ redox $2S_2O_3^{2-} + I_2 \rightarrow 2I^- + S_4O_6^{2-}$
6c(i)	10	The total volume of the iodide solution and water always adds up to 25cm³. This means the volume of water to be added to 15cm³ of iodide solution must be 10cm³.
6c(ii)	One answer from:	To keep the concentration (of the reactants) constant | Adding water will { change / affect / dilute / decrease } the concentration
6d(i)	Doubling concentration doubles reaction rate	When concentration = 0.01mol l^{-1} rate = 0.025s^{-1} When concentration = 0.02mol l^{-1} rate = 0.050s^{-1} Doubling concentration Doubles rate of reaction
6d(ii)	16.1	When concentration = 0.025mol l^{-1} then rate = 0.062s^{-1} $\text{rate} = \frac{1}{\text{time}} \quad \therefore \text{time} = \frac{1}{\text{rate}} = \frac{1}{0.062} = 16.1\text{s}$
6d(iii)	Answer to include:	1st mark: More particles/ions (in a given volume) 2nd mark: More successful collisions (and the faster the reaction)

7a	1129.4	ΔH = ΣBond enthalpies for bonds broken - ΣBond enthalpies for bonds formed $-107.6 = (1\times C{\equiv}O + 1\times Cl{-}Cl) - (2\times C{-}Cl + 1\times C{=}O)$ $-107.6 = (C{\equiv}O + 1\times243) - (2\times338 + 1\times804)$ $-107.6 = C{\equiv}O + 243 - (676 + 804)$ $-107.6 = C{\equiv}O + 243 - 1480$ $-107.6 = C{\equiv}O - 1237$ $C{\equiv}O = -107.6 + 1237$ $C{\equiv}O = 1129.4$
7b	Answer to include:	1st mark: Lower temperature/cooling favours exothermic forward reaction 2nd mark: Cooling below 100°C removes $H_2O_{(g)}$ from equilibrium as $H_2O_{(l)}$
7c(i)	Diagrams showing:	workable apparatus for production of CO_2 showing hydrochloric acid and calcium carbonate (1mark)  Apparatus showing gas being passed through sealed container of sodium hydroxide (1mark) 
7c(ii)	16.9	atom economy $= \frac{\text{mass of useful products}}{\text{total mass of reactants}} \times 100 = \frac{(1\times28.0)}{(1\times65.4)+(1\times100.1)}\times100 = \frac{28}{165.5}\times100 = 16.9\%$
8a(i)	198.4	64% Yield ⟷ 100g tin 100% Yield ⟷ 100g tin $\times \frac{100}{64}$ =156.25g tin **n**o. of mol $= \frac{\textbf{mass}}{\textbf{gfm}} = \frac{156.25}{118.7}$ = 1.32mol tin $SnO_2 + C \longrightarrow Sn + CO_2$ 1mol → 1mol 1.32mol → 1.32mol **m**ass = **n**o. of mol x **gfm** = 1.32 x 150.7 = 198.4g
8a(ii)	1 mark for calculation showing carbon is in excess 1 mark for 0.083mol	**gfm** SnO_2 = 150.7g mass = 25.2g **n**o. of mol $= \frac{\textbf{mass}}{\textbf{gfm}} = \frac{25.2}{150.7}$ = 0.167mol $SnO_2 + C \longrightarrow Sn + CO_2$ 1mol 1mol 0.167mol 0.167mol (required) **gfm** C = 12.0g mass = 3.0g **n**o. of mol $= \frac{\textbf{mass}}{\textbf{gfm}} = \frac{3.0}{12.0}$ = 0.25mol (available) ***C* is in excess** as more *C* available (0.25mol) than is required (0.0167mol) *C* leftover = 0.25mol - 0.167mol = 0.083mol
8a(iii)	49.9	**M**olar **V**olume = 80 l mol^{-1} **V**olume = 26.5 litres **n**o. of mol $= \frac{\textbf{Volume}}{\textbf{Molar Volume}} = \frac{26.5\text{ litres}}{80\text{ litres mol}^{-1}}$ = 0.331 mol $SnO_2 + C \longrightarrow Sn + CO_2$ 1mol → 1mol 0.331mol → 0.331mol **gfm** SnO_2 = 150.7g **m**ass = **n**o. of mol x **gfm** = 0.331 x 150.7 = 49.9g

8b(i)	filtration	Solid magnesium hydroxide can be removed from calcium chloride solution by filtration
8b(ii)	One answer from:	Seawater is a free/cheap raw material; Sell calcium chloride as by-product
9a(i)	Propagation	**Step** / **Reactants** (before Arrow) → **Products** (after Arrow) Initiation: No free radicals on Reactant Side → Free radicals on Product Side Propagation: Free Radicals found on both sides of arrow Termination: Free radicals on Reactant Side → No free radicals on Product Side
9a(ii)	Unpaired electron	A free radical is a particles containing an unpaired electron.
9b	Denaturing	Proteins are denatured when the shape of the protein is permanently changed often due to a rise in temperature or a change in pH.
9c	One answer from:	plastics; food products; cosmetics; or any other reasonable answer
9d(i)	Oxygen/gas is escaping	The gas produced in the chemical reaction is able to escape the test tube as there is no stopper and the balance will record this loss of mass.
9d(ii)	Two answers from:	concentration of hydrogen peroxide; temperature; mass / volume / surface area / size of potato discs
9d(iii)	One answer from:	As pH increases { the rate increases up to pH=10 / the rate reaches a maximum at pH=10 / the rate reaches an optimum at pH=10 } (and then decreases) or As the pH increases the rate increases and then decreases
10a(i)		 Amide link (also called peptide link in proteins only)  Ester link
10a(ii)	One answer from:	Bromine solution would { quickly / rapidly } decolourise with { CAP / capsaicin } Bromine solution will decolourise with { CAP / capsaicin } but not with { DHC / dihydrocapsaicin } Bromine solution not decolourise with { DHC / dihydrocapsaicin } but will with { CAP / capsaicin }
10b	One answer from:	Capsaicinoids do not dissolve in water; Capsaicinoids are insoluble in water; Capsaicinoids are hydrophobic; Capsaicinoids are non-polar
10c	135	CAP has retention time of 2.25 minutes = 135seconds
10d	3	0.025% of 60g tube = $\frac{0.025}{100} \times 60g$ = 0.015g CAP 1000g CAP ⟷ £1930.32 0.015g CAP ⟷ £1930.32 × $\frac{0.015g}{1000}$ = £0.029 = 2.9p = 3p

<table>
<tr>
<td>10e</td>
<td>$C_6H_8O_6$
↓
$C_6H_6O_6 + 2H^+ + 2e^-$</td>
<td><u>Step 1</u>: Write down main species in reaction
$C_6H_8O_6 \rightarrow C_6H_6O_6$
<u>Step 2</u>: Balance all atoms other than O or H
$C_6H_8O_6 \rightarrow C_6H_6O_6$
<u>Step 3</u>: Balance O atoms by adding H_2O to the other side
$C_6H_8O_6 \rightarrow C_6H_6O_6$
<u>Step 4</u>: Balance H atoms by adding H^+ to the other side
$C_6H_8O_6 \rightarrow C_6H_6O_6 + 2H^+$
<u>Step 5</u>: Balance charge by adding electrons to the most positive side
$C_6H_8O_6 \rightarrow C_6H_6O_6 + 2H^+ + 2e^-$</td>
</tr>
<tr>
<td>11</td>
<td>Open Question Answer:
<table><tr><td>Mark</td><td>0</td><td>1</td><td>2</td><td>3</td></tr><tr><td>%</td><td>57%</td><td>23%</td><td>14%</td><td>7%</td></tr></table></td>
<td><table><tr><th>3 mark answer</th><th>2 mark answer</th><th>1 mark answer</th></tr><tr><td>Demonstrates a <u>good</u> understanding of the chemistry involved. A good comprehension of the chemistry has provided in a logically correct, including a statement of the principles involved and the application of these to respond to the problem.</td><td>Demonstrates a <u>reasonable</u> understanding of the chemistry involved, making some statement(s) which are relevant to the situation, showing that the problem is understood.</td><td>Demonstrates a <u>limited</u> understanding of the chemistry involved. The candidate has made some statement(s) which are relevant to the situation, showing that at least a little of the chemistry within the problem is understood.</td></tr></table></td>
</tr>
<tr>
<td>12a</td>
<td>2,7-dimethyloctane</td>
<td></td>
</tr>
<tr>
<td>12b</td>
<td></td>
<td></td>
</tr>
<tr>
<td>12c</td>
<td>
or
</td>
<td></td>
</tr>
</table>